BTech 6th semester Syllabus

Wireless Networks	
Course Code: BCS 302 Contact Hours: L-3 T-0 P-2 Course Category: DCC	Credits: 4 Semester: 6

Introduction: This course is about teaching of the fundamental concepts of wireless networks and has a basic knowledge of the different types of ad-hoc networks and underlying protocols. Course will provide the understanding of the architecture of wireless networks for its various application setups.

Course Objective:

- To introduce the basics of wireless networks
- To make them familiar to the challenges involved in wireless networks with respect to wired networks.
- To introduce various types of wireless networks, i.e cellular networks, Bluetooth, Ad hoc networks, wireless mesh networks and wireless sensor networks.

Pre-requisite: Students should have basic knowledge of wireless communication and computer networks.

Course Outcome: Upon successful completion of this course, students will be able to:
CO1 Understand the underlying technologies of wireless networks.
CO2 Understand of the existing wireless protocols for MAC layer, Network layer and transport layer.
CO3 Understand the concepts of ad hoc networks and the design / performance issues in wireless local area networks and wide area networks.
CO4 Understand the function of the node architecture and use of sensors for various applications.
CO5 Understand the implementation strategies of distributed computing and wireless network protocols.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I	12 hours
Introduction: Introduction to wireless networks, Ad-hoc networks – definition, characteristics features, applications. Characteristics of Wireless channel, Adhoc Mobility Models: - Indoor and outdoor models. **MAC Protocols:** design issues, goals and classification. Contention based protocols, IEEE Standards: 802.11, 802.15.	
UNIT II	**10 hours**
Network Protocols: Routing Protocols: Design issues, Proactive Vs reactive routing protocols, ~~Unicast routing protocols, Multicast routing protocols,~~ hybrid routing protocols, Energy aware routing protocols, Hierarchical Routing protocols ~~**Transport Layer**: Issues in designing Transport Layer, Transport layer classification, Ad-hoc transport protocols.~~	
UNIT III	**10 hours**
Wireless Mesh Networks: Necessity for Mesh Networks, MAC enhancements, IEEE 802.11s Architecture, Opportunistic Routing, Heterogeneous Mesh Networks, Vehicular Mesh Networks	
UNIT IV	**10 hours**
Wireless Sensor Networks: Introduction, Sensor Network architecture, Data Dissemination, Data Gathering, Location discovery, Quality of Sensor Networks, Sensor Network Platforms and Tools, Energy Efficient Approaches	

Text Books	
1	C. Siva Ram Murthy and B. S. Manoj, Ad hoc Wireless Networks Architectures and Protocols, 2nd edition, Pearson Education, 2004.
2	Holger Karl & Andreas Willig, " Protocols and Architectures for Wireless Sensor Networks”, John Wiley, Student Edition (Indian), 2016.
Reference Books	
1	Perkins, Charles, “Ad hoc networking”, 1st Edition, Pearson Education India, 2008.
2	R. Hekmat, “Ad-hoc Networks: Fundamental Properties and Network Topologies”, Springer, 1st Edition, 2006
3	C. K.Toh, “Adhoc Mobile Wireless Networks”, Pearson Education; 1st edition, 2015.
4	W. Dargie and C. Poellabauer, “Fundamentals of Wireless Sensor Networks –Theory and Practice”, Wiley, 2010.

Microprocessor and Interfacing	
Course Code: BCS 304 **Contact Hours:** L-3 T-0 P-2 **Course Category**: DCC	Credits: 4 Semester: 6

Introduction: This course provides the complete description of architecture and instruction set of microprocessors. It also deals with assembly language programming and concept of Interfacing with various I/O devices and microprocessors. It covers all concepts of microprocessors for the development of real time solutions. This course also helps to understand the difference between 8086 and other advanced microprocessors.

Course Objective:

- Ability to Demonstrate an understanding of the design of the functional units of a microprocessor.
- Explain the instruction set, instruction formats, Addressing modes of the microprocessor and ability to Recognize and manipulate representations of numbers stored in the microprocessor and perform Basic Arithmetic Operations.
- Ability to analyze memory hierarchy and its impact on computer Cost/performance.
- Understanding and interfacing communication modules with a processor for various real time applications.

Pre-requisite: Computer Organisation

Pre-requisite: Basic electronics

Course Outcome: Upon successful completion of this course, students will be able to:
CO1 Understand the basic concepts of microprocessors
CO2 Understand the instruction set of 8086.
CO3 Apply the knowledge of assembly language to solve various problems.
CO4 Grasp an understanding of various peripheral device interfaces with 8086.
CO5 Design and implement various interfaces in real life different applications

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT-I	10 Hours
Introduction and architecture: Introduction to microprocessors, classification, basic architecture and its applications. History of microprocessors, Brief Introduction to 8085 Microprocessor, , 8086 architecture, BIU and EU, Register Organization, Pin diagram, Memory Segmentation	
UNIT-II	**12 Hours**
Assembly Language Programming of 8086: Instruction set of 8086, Instruction execution time, Timing Diagram, fetch cycle, execution cycle. Instruction format, Addressing modes, Instruction Types (data transfer instruction, arithmetic instructions, branch instruction, NOP & HLT instructions, flag manipulation instruction, logical instruction, shift and rotate instruction, String instructions), Assembler directions and operators.	
UNIT-III	**10 Hours**
Introduction to Multiprogramming: System Bus Structure ,Multiprocessor configurations, Coprocessor, Closely coupled and loosely Coupled configurations, Introduction to advanced processors, 80186, 80286, 80386 **Interfacing:** Memory interfacing to 8086, Interrupt structure of 8086, Vector interrupt table, Interrupt service routine. Interfacing Interrupt Controller 8259	
UNIT-IV	**10 Hours**
Interfacing with advanced devices: I/O Interface: 8255 PPI various modes of operation and interfacing to 8086. Communication Interface: Serial data transfer schemes. 8251 USART architecture and interfacing, DMA , Interfacing DMA Controller 8257 to 8086.	

Text Books	
1	A.K. Ray and K.M. Bhurchandi, “Advanced Microprocessors and Peripherals “, McGraw Hill Education; 3rd edition (2017)
2	Douglas V Hall, SSSP Rao, “Microprocessor and Interfacing”, McGraw Hill Education; 3rd edition (2017)
3	Y.C. Liu and A. Gibson, “Microcomputer systems-The 8086 / 8088 Family”,Pearson Education India; 2nd edition (2015)
Reference Books	
1	Barry B. Brey, “The Intel Microprocessor, Architecture, Programming and Interfacing “ , Pearson Education India; 8th edition (2008)
2	Kenneth J Ayala, “The 8086 Microprocessor: Programming & Interfacing the PC” , Cengage Learning; 1st edition (2007)
3	B. Ram, ‘Fundamentals of Microprocessors and Microcomputers’, Dhanpat Rai Publications. latest edition, 2012

Compiler Design	
Course Code: BCS 306 **Contact Hours**: L-3 T-0 P-2 **Course Category**: DCC	Credits: 4 Semester: 6

Introduction: This course provides the complete description about inner working of a compiler. This course focuses mainly on the design of compilers and optimization techniques. It also includes the design of Compiler writing tools. This course also aims to convey the language specifications, use of regular expressions and context free grammars behind the design of compiler.

Course Objective:

- To Introduce the concepts of language translation and compiler design
- To impart the knowledge of practical skills necessary for constructing a compiler

Pre-requisite: Programming in C

Course Outcome: Upon successful completion of this course, students will be able to:

- Understand the concepts and different phases of compilation with compile time error handling.
- Represent language tokens using regular expressions, context free grammar and finite automata and design lexical analyzer for a language.
- Compare top down with bottom up parsers, and develop appropriate parser to produce parse tree representation of the input.
- Design a compiler for a small subset of C language.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT-I	10 Hours
Introduction to compilers – Analysis of the source program, Phases of a compiler, Grouping of phases, compiler writing tools– bootstrapping. Case study: MiniC (A small subset of C language) **Lexical Analysis-**The role of Lexical Analyzer, Input Buffering, Specification of Tokens using Regular Expressions, Review of Finite Automata, Recognition of Tokens Case study: Lexical Analysis for MiniC **Syntax Analysis:** Review of Context-Free Grammars – Derivation trees and Parse Trees, Ambiguity.	
UNIT-II	**12 Hours**
Top-Down Parsing: Recursive Descent parsing, Predictive parsing, LL(1) Grammars**.** **Bottom-Up Parsing**: Shift Reduce parsing – Operator precedence parsing (Concepts only). LR parsing – Constructing SLR parsing tables, Constructing Canonical LR parsing tables and Constructing LALR parsing tables. Case study: Syntax analysis for MiniC	
UNIT-III	**10 Hours**
Syntax directed translation: Syntax directed definitions, Bottom- up evaluation of S- attributed definitions, L- attributed definitions, Top-down translation, Bottom-up evaluation of inherited attributes. **Type Checking**: Type systems, Specification of a simple type checker. **Run-Time Environments**: Source Language issues, Storage organization, Storage allocation strategies.	
UNIT-IV	**10 Hours**
Intermediate Code Generation (ICG): Intermediate languages – Graphical representations, Three Address code, Quadruples, Triples. Assignment statements, Boolean expressions. **Code Optimization**: Principal sources of optimization, Optimization of Basic blocks. **Code generation**: Issues in the design of a code generator. A simple code generator. Case study: MiniC Code Generator for the MiniC Architecture	

Text Books

1	Aho A., M. S Lam, R. Sethi and D Ullman, "Compilers – Principles Techniques and Tools", Pearson Education India; 2nd edition (2013)
2	K. C. Louden, "Compiler Construction – Principles and Practice", Cengage Learning Indian Edition, 2006.

Reference Books

1	A. I Hollub, Compiler Design in C, Pearson Education India; 1st edition (2015)
2	AW Appel, M Ginsburg, "Modern Compiler Implementation in C", Cambridge University Press, 2004.
3	K. Muneeswaran, "Compiler Design" ,Oxford University Press, Illustrated edition (2012)

Multimedia Technologies	
Course Code: BCS 308 **Contact Hours**: L-3 T-0 P-2 **Course Category**: DCC	Credits: 4 Semester: 6

Introduction: This course is an introduction to the technologies encompassing digital multimedia including images, videos, audios, web pages etc. that allow users to use and communicate with the multimedia over the internet. After learning this course, the student should be able to understand the basic concepts and technologies related to the multimedia, web, hardware and software programs needed to develop and deploy such computer based interactive multimedia applications.

Course Objective:

- To introduce the basic concepts of multimedia data representations and data compression
- To introduce multimedia architecture and databases
- To introduce the various multimedia applications.

Pre-requisite: Programming Concepts, Graphics.

Course Outcome: Upon successful completion of this course, students will be able to:

CO1 To understand the importance of multimedia objects for web application development

CO2 To understand and apply the compression techniques for various multimedia objects

CO3 To understand and study multimedia database systems and content-based information retrieval systems

CO4 To understand the various multimedia applications and latest trends

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT-I	10 Hours
Introduction: Brief introduction about multimedia concepts. Multimedia objects. Text, graphics, sound, video, images, internet and web, world wide web, html, XML, Angular JS, CSS. Hardware and software requirements. Multimedia software tools, hardware. Concept of Non- Temporal and Temporal Media. Basic Characteristics of Non-Temporal Media, Hypertext and Hypermedia. Presentations, Synchronization, Events, Scripts and Interactivity.	
UNIT-II	**10 Hours**
Data representation and compression: Data representation, Basic concepts of Compression, basic compression techniques, video audio data compression techniques, image format. Huffman coding, Shannon Fano algorithm, Huffman algorithms, adaptive coding, arithmetic coding higher order modeling, finite context modeling, dictionary based compression, sliding window compression, LZ77, LZW compression, compression ratio loss less and lossy compression.	
UNIT-III	**10 Hours**
Multimedia system architecture & database: Introduction to Multimedia PC/Workstation Architecture, Characteristics of MMX instruction set, I/O systems: Overview of USB port and IEEE 1394 interface, Operating System Support for Multimedia, Multimedia Database Design, Content Based Information Retrieval: Text Retrieval, Image Retrieval, Video Retrieval, Overview of MPEG-7, and design of video-on-Demand Systems.	
UNIT-IV	**10 Hours**
Multimedia & its Applications: Speech (digital audio concepts, sampling variables, loss; loss less and silence compression), Images (multiple monitors, bitmaps, vector drawing, lossy graphic compression, JPEG compression, Zigzag coding), Multimedia network and QoS support, multimedia wireless networks, heterogeneous network, multimedia applications. Latest trends in multimedia and case study related to multimedia technologies.	

Text Books	
1	D. Hillman, “Multimedia Technology & Applications”, Galgotia Publications. First edition 2008.
2	N. Chapman & J. Chapman, “Digital Multimedia”, Wiley Pub. Third edition 2009.
Reference Books	
1	O. Marsh. “Multimedia Technology and Applications.” Larsen and Keller Edu. 2017.
2	R. Agrawal, “Multimedia Systems”, Excel Books. 2013.
3	D.P. Mukherjee, “Fundamentals of Computer Graphics and Multimedia”, PHI 2004.
4	Z. Li & M. S. Drew. “Fundamentals of Multimedia.” Prentice Hall India Learning Private Limited (2005)

Human Computer Interaction	
Course Code: BCS 310 **Contact Hours**: L-3 T-1 P-0 **Course Category**: DEC	Credits:4 Semester:5

Introduction: HCI is a multidisciplinary course focusing on design of computer technology interaction between humans particularly users, and computers. HCI has expanded and evolved and now covers several forms of information technology design.

Course Objective:

- To Impart knowledge and understanding about human cognition and human perspective with respect to computers.
- To Introduce User Interface and Graphical User Interface elements
- To introduce components and software tools for Human computer interaction.

Pre-requisite: Computer fundamentals

Course Outcome: Upon successful completion of this course, students will be able to:

CO1 To understand the importance of good user interface design and the difference between UI and GUIs

CO2 To understand and apply the design process of UIs

CO3 To understand and apply screen-based UI principles

CO4 To apply the UI design and interaction-based device components to UI designing

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped classroom teaching will be adopted.

UNIT I	**10 Hours**
Introduction: Importance of user interface, definition, importance of good design, Benefits of good design, A brief history on screen design. **Graphical User Interface:** Popularity of Graphics, the concept of direct manipulation, graphical system, characteristics, web user interface popularity, characteristics-principles of user interface.	
UNIT II	**11 Hours**
Design process: Human interaction with computers, importance of human characteristics, human considerations, human interaction speeds, understanding business junctions. Limitations and constraints of mobile and hand-held platforms and issues related to network, power, memory and UI responsiveness required with reference to other computing platforms.	
UNIT III	**11 Hours**
Screen Designing (UI Design): Design goals, screen planning and purpose, organizing screen elements, ordering of screen data and content, screen navigation and flow, visually pleasing composition, amount of information, focus and emphasis, presenting information simply and meaningfully, information retrieval on web, statistical graphics, technological consideration in interface design. **Windows:** Navigation schemes selection of window, selection of devices based and screen-based controls.	
UNIT IV	**10 Hours**
Components and Software Tools: Components: text and messages, Icons and increases, multimedia, colors, user problems. **Software tools:** Specification methods interface, building tools. **Interacting Devices:** Keyboard and function keys, touch screen technology, pointing devices, speech recognition digitization and generation, image and video displays (LED,LCD,HD displays), voice activated commands and its implementation	

Text Books	
1	W. O Galitz, “The essential guide to user interface design”, Wiley, Third Edition, 2007.
2	B. Shneidermann, “Designing the User Interface: Strategies for Effective Human-Computer Interaction”, 5th Edition,Pearson Education ,2014.
Reference Books	
1	A. Dix, J. Fincay, G. Goryd, Abowd, R. Bealg, “ Human Computer Interaction” , 3rdEdition, Pearson Education,2004.
2	Sharp ,Rogers, Preece, “Ineraction Design Beyond Human-Computer Interaction”, 5th Edition, Wiley, 2019
3	D. Stone, C. Jarrett, M. Woodroffe, and S. Minocha, “User Interface Design and Evaluation(interactive technologies) , Morgan Kaufmann; Revised of "The Miracle of Col ed. edition (2005)

Advanced Computer Architecture	
Course Code: BCS 312 **Contact Hours**: L-3 T-1 P-0 **Course Category**: DEC	Credits: 4 Semester: 5

Introduction: This course provides the complete description about the advancements in Computer Architecture. After exploiting the full capacity of execution of uniprocessor system, the speed is enhanced with using multiprocessor and other concepts like pipelining. The algorithms also need to be parallelised for achieving highest speed. This course aims at teaching the complete concepts about the changes in bus system, memory, placements and interconnection of different processors etc.

Course Objective:

- To introduce different concepts for enhancing speed of computation
- To make students learn about different computer architectures.
- To introduce the concepts of parallel algorithms and parallel programming

Pre-requisite: A course on computer organisation, microprocessor, and computer architecture

Course Outcome: Upon successful completion of this course, students will be able to:

CO1 Understand the concept of parallel architecture and Analyse performance metrics
CO2 Understand the concept of pipelining for achieving higher speed.
CO3 Understand and Analyse different parallel architectures.
CO4 Understand the concepts of parallel algorithms and parallel programming

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I	12 Hours
Introduction & Fundamentals: The concept of computer Architecture: Interpretation of concept of computer architecture at different level abstraction, Multi-level hierarchical frame work, description of computer architecture, Introduction to parallel processing: Basic concept, types of level of parallelism, classification of parallel architecture, Basic parallel techniques, relationship between language and parallel architecture. Principles of scalable performance: Performance Metrics and Measures, Speedup Performance Law, Scalability Analysis & approaches, Processor and memory hierarchy: Design Space of Processor, ISA, CISC & RISC, Memory Hierarchy Technology, Virtual Memory Technology	
UNIT II	**10 Hours**
Instruction Level Parallel Processor (Parallelism): Pipelined Processors: Basic concept, ILP: Basics, Exploiting ILP, Limits on ILP, design space of pipelines, performance of pipeline, reservation table, And DLX Case Study. VLIW architecture, Superscalar Processor: Super Scalar and super-pipeline Design, A case study of ARM 64-bit processor.	
UNIT III	**12 Hours**
Data parallel Architecture and MIMD architectures: SIMD Architecture: Design space, fine grain SIMD architecture, coarse grain SIMD architecture, Associative and Neural Architecture, Systolic Architecture, Vector Architectures: Wordlength, vectorization, pipelining, and vector instruction format**. Thread and Process Level Parallel Architecture (MIMD Architecture):** Multi-threaded Architecture: Design space, computational model, Data flow architecture, hybrid multi shared architecture Distributed memory MIMD, Architecture: Design space, interconnection networks, topology, fine grain system, medium grain system, coarse grain system, Cache Coherence and synchronisation Mechanism Shared memory MIMD Architecture.	
UNIT-IV	10 Hours
Parallel Algorithm and Programming: MPI: Basics of MPI Open MP: Open MP Implementation in 'C', Directives: Conditional Compilation, Internal Control Variables, Parallel Construct, Work Sharing Constructs, Combined Parallel Work-Sharing constructs, Master and Synchronisation Constructs POSIX thread: IEEE POSIX Threads: Creating and Exiting Threads, Simultaneous Execution of threads.	

Text Books

1	D. Sima, T. Fountain , P. Karsuk , "Advanced Computer Architectures: A Design Space Approach", Pearson Education India; 1st edition, 2002.
2	K. Hwang , N. Jotwani , "Advance Computer Architecture : Parallelism, Scalability, Programmability", McGraw Hill Education; 3rd edition, 2017.

Reference Books

1	Quinn, "Parallel Programming in C with MPI and Open MP", McGraw Hill Edu.; 1st Ed. 2017.
2	J. P. Hayes, "Computer Architecture and Organization", McGraw Hill Education; 3rd ed. 2017.

Cloud Computing	
Course Code: BIT 304 **Contact Hours**: L-3 T-0 P-2 **Course Category**: DCC	Credits: 4 Semester: 6

Introduction: Cloud computing is a scalable services provider platform that provides on-demand and pay per use computing service for various types of shared pool of resources such as memory, servers, storage, networking, software, database, applications designing etc., with the help of internet. This course will introduce various aspects of cloud computing including fundamentals of cloud computing, load balancing techniques, security challenges, case studies and industrial applications of cloud computing. This will help students to use and explore the cloud computing platforms.

Course Objective:

- Learn the use Cloud Services and Cloud Deployment models.
- Understand the concept of virtualization in cloud computing.
- Gaining confidence in resource management and load balancing algorithms.
- To gain the confidence of security attacks and their provisions at various levels of cloud computing.

Pre-requisite: Basic understanding of Operating System, Internet, Parallel and Distributed Computing.

Course Outcome: Upon successful completion of this course, students will be able to:

CO1: To articulate key concepts of cloud computing and computing techniques, strength and limitations of cloud computing with possible application domains.

CO2: To identify the architecture and infrastructure of cloud computing including SaaS, PaaS, IaaS, public cloud, private cloud and hybrid cloud.

CO3: To interpret various data, scalability and cloud services to acquire efficient database for cloud storage.

CO4: To explain the core issues of cloud computing such as security, privacy and interoperability and deal with controlling mechanism for accessing sage cloud service.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I	**10 hours**
Cloud Computing Fundamentals: Introduction of cloud computing, History of cloud computing, Trends in computing, Grid computing, Cluster computing, Distributed computing, Utility computing, Fog computing, NIST definition and characteristics of cloud computing, Cloud as green and smart, Cloud as IaaS, PaaS, SaaS, BPaaS and HaaS, SPI framework, SPI vs. traditional IT Model, Cloud deployment models, Benefits and challenges.	
UNIT II	**10 hours**
Virtualization and Cloud Architecture: Virtualization concept, Resource virtualization, Server virtualization, Storage virtualization and Network virtualization, Storage Network Design: Architecture of storage, Analysis and planning, Storage models, Cloud optimized storage, Virtual Box and Microsoft Hyper-V.	
UNIT III	**10 hours**
Cloud Security: Web services, Web 2.0, Web OS, Security challenges and preventive measures: Infrastructure layer, Network layer and Application layer of cloud computing architecture, Security models in cloud, Resource management in cloud computing, Static and dynamic load balancing in cloud computing, Identity access management and Trust in cloud computing, Thin client.	
UNIT IV	**10 hours**
Cloud providers and case studies: Amazon EC2, Amazon EC service level agreement and recent developments, GoGrid, Salesforce.com, Force.com, Google App Engine, Rackspace, Government of India Cloud, IBM cloud, Eucalyptus cloud, Analysis of Case Studies when deciding to adopt cloud computing architecture.	
Text Books	
1	B. Sosinsky, "Cloud Computing Bible", 1^{st} Edition, Wiley-India, 2011/ Latest Edition.
3	Thomas Erl, Zaigam Mahmood, Ricardo Puttini, "Cloud Computing Concepts, Technology & Architecture", 1^{st} Edition, Pearson India, 2013/ Latest Edition.
Reference Material	
1	A. Shawish and M. Salama, "Cloud computing: paradigms and technologies." In Inter-cooperative collective intelligence: Techniques and applications, Springer, 2014/ Latest Edition.
2	M. Miller, "Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online", 1^{st} Edition, Pearson Education India, 2008/ Latest Edition
3	https://swayam.gov.in/course/4413-cloud-computing
4	https://nptel.ac.in/noc/courses/noc20/SEM1/noc20-cs20/

Internet of Things	
Course Code: BIT 310 **Contact Hours**: L-3 T-0 P-2 **Course Category**: DEC	Credits: 4 Semester: 6

Introduction: Internet of Things (IoT) is the next big idea in technology and has gained prominence with the ever-increasing connected devices, sensor systems and capability of computing resources. This course is designed to initiate the widest possible group of students to the field of IoT and will be comprehensive in its scope. This course supplies in-depth content that puts the theory into practice. The course will start with a basic introduction to IoT and take the students through an IoT solution case study.

Course Objective:

- Impart understanding of various building blocks and working of state-of-the-art IoT systems.
- Learn the basic issues, policy and challenges in the Internet and understand the cloud and internet environment.
- Design and program own IoT devices by using real IoT communication protocols.
- Analyze the data generated from the IoT devices.

Pre-requisite: Design and Analysis of Algorithms, Data Structures and Algorithms and Computer Networks

Course Outcome: After completion of this course, the students will be able to:
CO1 Develop smart IoT Applications using smart sensor devices and cloud systems.
CO2 Analyze the protocol Stack for IoT in order to address the issues related to heterogeneous devices and networks.
CO3 Design IoT system specific secure protocols.
CO4 Understand uses and risks related to IoT devices.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I	10 hours
Introduction: Definition, Functional requirements, Characteristics, Foundations, architectures, challenges and issues, Physical design of IoT, Logical design of IoT, Web 3.0 of IoT, IoT World Forum (IoTWF) and Alternative IoT models, IoT Communication Models, IoT in Global Context, Real world scenarios, Different Areas, Examples Trends in the Adaption of the IoT (Cloud Computing, Big Data Analytics, Concepts of Web of Things, Concept of Cloud of Things with emphasis on Mobile Cloud Computing, Smart Objects).	
UNIT II	**10 hours**
Components in IoT: Control Units, Sensors, Communication modules, Power Sources, Communication Technologies, RFID, Bluetooth, Zigbee, Wi-fi, RF links, Mobile Internet, Wired Communication**; IoT Protocol and Technology:** RFID, NFC, Wireless Networks, WSN, RTLS , GPS, Agents , Multi – Agent Systems, IoT Protocols: M2M, BacNet, ModBus, Bluetooth, Wi-Fi, ZigBee; **Web of Things (WoT):** WoT vs. IoT, Architecture; **Cloud of Things (CoT):** Grid/SOA and Cloud Computing, Standards, Cloud Providers and Systems, Architecture.	
UNIT III	**10 hours**
Data Analytics for IoT: Introduction, Machine Learning, Big Data Analytics Tools and Technology, Apache Hadoop, Using Hadoop MapReduce for Batch Data Analysis, Apache Oozie, Apache Spark, Apache Storm, Apache Kafka, Edge Streaming Analytics and Network Analytics, Xively Cloud for IoT, Using Apache Storm for Real-time Data Analysis, Structural Health Monitoring Case Study, **Tools for IoT**: Chef, Chef Case Studies, Puppet, Puppet Case Study – Multi-tier Deployment, NETCONF-YANG Case Studies, IoT Code Generator.	
UNIT IV	**10 hours**
Domain specific applications of IoT: Home automation, Industry applications, Surveillance applications, Smart Homes, Ambient Assisted Living, Intelligent Transport, Other IoT application: Use-Case Examples; **Developing IoT solutions**: Introduction to Python, Introduction to different IoT tools, Introduction to Arduino and Raspberry Pi Implementation of IoT with Arduino and Raspberry, Cloud Computing, Fog Computing, Connected Vehicles, Data Aggregation for the IoT in Smart Cities, Privacy and Security Issues in IoT.	

Text Books

1	A. Bahga, V. Madisetti, "Internet of Things: A Hands-on Approach", 1st Edition, Universities Press, 2015/ Latest Edition.
2	R. Kamal, "Internet of Things: Architecture and Design Principles", 1st Edition, McGraw Hill Education private limited, 2017/ Latest Edition.

Reference Material

1	D. Uckelmann, M. Harrison, "Architecting the Internet of Things", 1st Ed., Springer, 2011/ Latest Edition.
2	O. Hersent, D. Boswarthick, O. Elloumi, "The Internet of Things – Key applications and Protocols", 2nd Edition, Wiley, 2012/ Latest Edition.

Computer Graphics	
Course Code: BCS 314 **Contact Hours**: L-3 T-0 P-2 **Course Category**: DEC	Credits: 4 Semester: 6

Introduction: The subject Computer Graphics introduces basic concepts of graphics, output primitives, transformations, projections, curve and surface generation methods and shading algorithms.

Course Objective:

- To introduce the basic concepts of computer graphics
- To introduce the concepts of 2D/3D transformations
- To introduce the concepts of curve generation and hidden surface detection.

Pre-requisite: Basic mathematics

Course Outcome: Upon successful completion of this course, students will be able to:
CO1 Understand basic concepts of computer graphics and its applications.
CO2 Use the 2D/3D transformation and projection concepts in various projects
CO3 Understand concepts of curve generation and hidden surface detection.
CO4 Develop various application of computer graphics

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I	10 hours
Introduction to computer graphics: Introduction, Application of computer graphics, Video Display Devices, Raster Scan Systems, Random Scan Systems, Graphics Monitors and Work Stations, Input Devices, Hard Copy Devices, Graphics Software. Colour Models: RGB, HSV etc. Output primitives: DDA Line drawing algorithm, Bresenham's Line Drawing Algorithm, Mid-point circle algorithm, Mid-point Ellipse algorithms, filling algorithms, boundary fill and flood fill algorithms, scanline filling.	
UNIT II	**11 hours**
Transformations: Basic 2D Transformations, Matrix representations & Homogeneous Coordinates, Matrix Representations for basic 2D and 3D transformations (Translation, Scaling, Rotation), Composite Transformations, reflection and shear transformations, affine transformation, transformations between coordinate systems. Two-dimensional viewing: The viewing Pipeline, Window to view port coordinate transformation, Clipping Operations: Point Clipping, Line Clipping (Cohen Sutherland and Liang-Barsky), Polygon Clipping, Sutherland-Hodgeman polygon clipping, Wailer-Atherton polygon clipping, curve clipping, Text clipping.	
UNIT III	**10 hours**
Curves and Surfaces: Representation of surfaces, polygon meshes, plane equations, parametric cubic curves, Hermite Curves, Bezier Curves, 4 point and 5 point Bezier curves using Bernstein Polynomials, Conditions for smoothly joining curve segments, Bezier bi-cubic surface patch, B-Spline Curves, Cubic B-Spline curves using uniform knot vectors, Testing for first and second order continuities. Visible surface detection, Back Face Detection, Depth Buffer (Z-Buffer, A-Buffer) Method. Scan Line Method, Depth Sorting Method, Area Subdivision Method.	
UNIT IV	**11 hours**
Three-Dimensional Concepts: 3D Transformations, Parallel Projection and Perspective Projection. Shading and Illumination Model: Shading, Illumination Model for diffused Reflection, Ambient light, Specular Reflection Model, Reflection Vector. Shading Models, Flat shading, Gourard Shading, Phong Model. Case studies:Design case studies to perform 2D representations of lines and curves, perform 2D and 3D transformations on different objects.	

Text Books

1	D. Hearn and M. P. Baker, "Computer Graphics C version", Pearson Education; 2nd ed. (2014)
2	Z. Xiang & R. Plastock "Computer Graphics", Schaum's Series, McGraw Hill Edu. 2nd Ed (2006)

Reference Books

1	D. Rogers and J. Adams, "Mathematical Elements for Computer Graphics", McGraw Hill Education; 2nd edition (2017)
2	Foley, V. Dam, Feiner and Hughes, "Computer Graphics Principles & practice", Pearson Education India; 2nd edition (2002)

Principles of Management	
Course Code: HMC 302 Contact Hours: L-2 T-0 P-0 Course Category: HMC	Credits: 2 Semester: 6

Introduction: To give a preview of basics of management to engineering students, this course discusses about the basic nature of management and describes the functions of management, the specific roles of contemporary management, different approaches to designing organizational structures. This will help the students to understand the role of personality, learning and emotions at work, discover and understand the concept of motivation, leadership, power and conflict, understand the foundations of group behavior and the framework for organizational change and development.

Course Objective:

- To acquaint the students with the fundamentals of managing business
- To make them understand individual and group behavior at workplace so as to improve the effectiveness of an organization.
- The course will use and focus on Indian experiences, approaches and cases.

Pre-requisite: Communication skills

Course Outcome: After completion of the course, the students should be able to:
CO1 Understand the nature of management and describe the functions of management·
CO2 Understanding the specific roles of contemporary management.
CO3 Develop understanding of different approaches to designing organizational structures.
CO4 Understand the role of personality, learning and emotions at work.
CO5 Discover and understand the concept of motivation, leadership, power and conflict.
CO6 Understand the foundations of group behavior and the framework for organizational change and development.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I	7 hours
Introduction: Concept, Nature, Process and Significance of Management; Managerial levels, Development of Management Thought: Classical, Neo-Classical, Behavioral, Systems and Contingency Approaches.	
UNIT II	**7 hours**
Planning: Nature, Scope and Objectives of Planning; Types of plans; Planning Process; Organizing: Nature, Process and Significance; Principles of an Organization; Span of Control; Types of an Organization.	
UNIT III	**7 hours**
Staffing: Concept, Nature and Importance of Staffing. Motivating and Leading: Nature and Importance of Motivation; Types of Motivation; Leadership: Meaning and Importance; Traits of a leader.	
UNIT IV	**7 hours**
Controlling: Nature and Scope of Control; Types of Control; Control Process; Control Techniques– Traditional and Modern; Effective Control System.	

Text Books

1	S.P. Robbins, "Fundamentals Management: Essentials Concepts Applications", Pearson Education, 2014.
2	Gilbert, J.A.F. Stoner and R.E. Freeman, "Management", Pearson Education, 2014. H. Koontz, "Essentials of Management", McGraw Hill Education, 2012.

Reference Books

1	C. B. Gupta, "Management Concepts and Practices", Sultan
2	W. Ghillyer, "Management- A Real World Approach", McGraw Hill Education, 2010.
3	K. Mukherjee, "Principles of Management", McGraw Hill Education, 2012.

Marketing Management	
Course Code: HMC 304 **Contact Hours**: L-2 T-0 P-0 **Course Category**: HMC	Credits: 2 Semester: 6

Introduction: This course will build the basic concept of marketing and related concepts for the engineering students. It will provide an in-depth understanding to various elements of marketing mix for effective functioning of an organization. Students will learn some of the tools and techniques of marketing with focus on Indian experiences, approaches and cases.

Course Objective:

- To familiarize students with the marketing function in organizations.
- To equip the students with understanding of the Marketing Mix elements and sensitize them to certain emerging issues in Marketing.

Pre-requisite: Basic economics

Course Outcome: After completion of the course, the students should be able to
CO1 Understand the concept of marketing and related concepts.
CO2 An in-depth understanding to various elements marketing mix for effective functioning of an organization.
CO3 Learn some of the tools and techniques of marketing with focus on Indian experiences, approaches and cases.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I	**7 hours**
Introduction to Marketing: Nature, Scope and Importance of Marketing, Basic concepts, Marketing Environment.	
UNIT II	**7 hours**
Product: Product Levels, Product Mix, Product Strategy, Product Development, Product Lifecycle and Product Mix Pricing Decisions.	
UNIT III	**7 hours**
Place: Meaning & importance, Types of Channels, Channels Strategies, Designing and Managing Marketing Channel.	
UNIT IV	**7 hours**
Promotion: Promotion Mix, Push vs. Pull Strategy; Promotional Objectives, Advertising-Meaning and Importance, Types, Media Decisions, Promotion Mix, Personal Selling-Nature, Importance and Process.	

Text Books	
1	P. Kotler, P.Y. Agnihotri and E.U. Haque, "Principles of Marketing- A South Asian Perspective", Pearson Education, 2012.
2	T. Ramaswamy and S. Namkumar, "Marketing Management Global Perspective: Indian Context", McMillan, Delhi, 2013.
Reference Books	
1	R. Saxena, "Marketing Management", McGraw Hill Education, 2012.
2	C.W. Lamb, J.F. Hair, C. McDaniel, D. Sharma, "MKTG: a South Asian Perspective with Coursemate", Cengage Learning, 2016.
3	R. Winer, "Marketing Management", Pearson Education, 2012.

Financial Management	
Course Code: HMC 306 Contact Hours: L-2 T-0 P-0 Course Category: HMC	Credits: 2 Semester: 6

Introduction: Efficient Management of a business enterprise is closely linked with the efficient management of its finances. Accordingly, the objective of the course is to familiarize the engineering students with the basic fundamentals, principles and practices of financial decision-making in a business unit in the context of a changing, challenging and competitive global economic environment. The purpose of the course is to offer the students relevant, systematic, efficient and actual knowledge of financial management that can be applied in practice while making financial decisions and resolving financial problems.

Course Objective:
- To acquaint the students with the overall framework of financial decision-making in a business unit.
- To acquaint the students with the fundamentals of Financial Management
- To make them understand Decisions to be taken as a Finance Manager.
- The course will use and focus on Indian experiences, approaches and cases.

Pre-requisite: Basic economics

Course Outcome: Upon successful completion of the course, students will be able to:
CO1 Understand the overall role and importance of the finance function for decision-making.
CO2 Recommend whether and why a particular investment should be accepted or rejected by determining an appropriate investment criteria and projecting cash flows associated with corporate project evaluation.
CO3 Differentiate between the various sources of finance and their pros and cons.
CO4 Outline capital requirements for starting a business and management of working capital.
CO5 Analyse the complexities associated with management of cost of funds in the capital structure.
Apply the concepts of financial management to contemporary financial events.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

UNIT I	7 hours
Financial Management Definition, scope, objectives of Financial Management, Functions of a finance manager, Time value of money. Sources of Finance for different Organizations.	
UNIT II	**7 hours**
Capital Structure: Meaning of Capital Structure: Factors Determining Capital Structure. Cost of Capital: Concept, Importance and Classification.	
UNIT III	**7 hours**
Capital Budgeting: Concept, Importance and Appraisal Methods: Pay Back Period, Accounting, Rate of Return, Net Present Value Method (NPV), Profitability Index, and IRR. Capital Rationing.	
UNIT IV	**7 hours**
Working Capital Management: Operating cycle, Working Capital Estimation, Inventory Management: EOQ Problem.	

Text Books

1	M.Y. Khan and P.K. Jain, "Financial Management", McGraw Hill Education, 8th Edition, 2018.
2	I. M. Pandey, "Financial Management", Vikas Publishing House, 2015.

Reference Books

1	S. Kapil, "Financial Management", Pearson Education, 2012.
2	C. Prasanna, "Financial Management: Theory and Practice", McGraw Hill, 2017.
3	S.N. Maheshwari, "Financial Management: Principles and Practice", Sultan Chand, LN, 2019.

Human Resource Management	
Course Code: HMC 308 **Contact Hours**: L-2 T-0 P-0 **Course Category**: HMC	Credits: 2 Semester: 6

Introduction: This course focuses on issues and strategies required to select and develop manpower resources. The main objective of this course is to help the students to acquire and develop skill to design rational decisions in the discipline of human resource management.

Course Objective: The objective of this course is to make students familiar with the basic concepts of human resource management and people related issues.

- To enable the students to understand the HR Management and system at various levels in general and in certain specific industries or organizations.
- To help the students focus on and analyze the issues and strategies required to select and develop manpower resources.
- To develop relevant skills necessary for application in HR related issues.
- To enable the students to integrate the understanding of various HR concepts along with the domain concept in order to take correct business decisions.

Pre-requisite: Soft skills

Course Outcome: After completion of the course, the students should be able to:
CO1 Develop an understanding of the concept of human resource management and to understand its relevance in organizations.
CO2 Develop necessary skill set for application of various HR issues.
CO3 Analyze the strategic issues and strategies required to select and develop manpower resources.
CO4 Integrate the knowledge of HR concepts to take correct business decisions.

Pedagogy: The teaching-learning of the course would be organized through lectures, tutorials, assignments, projects/ presentations and quizzes. Faculty members strive to make the classes interactive so that students can correlate the theories with practical examples for better understanding. Use of ICT, web-based resources as well as flipped class room teaching will be adopted.

<table>
<tr><td colspan="2">UNIT I</td><td>7 hours</td></tr>
<tr><td colspan="3">Human Resource Management: Introduction to Concept and Functions of HRM, Role, Status and Competencies of HR Manager, HR Policies, Evolution of HRM. Emerging Challenges of Human Resource Management;</td></tr>
<tr><td colspan="2">UNIT II</td><td>7 hours</td></tr>
<tr><td colspan="3">Human Resource Planning: Human Resource Planning- Quantitative and Qualitative dimensions; Recruitment – Concept and sources; (E-recruitment, recruitment process outsourcing etc.); Selection – Concept and process; test and interview; placement induction. Job analysis – job description and job specification.</td></tr>
<tr><td colspan="2">UNIT III</td><td>7 hours</td></tr>
<tr><td colspan="3">Training and Development: Concept and Importance; Identifying Training and Development Needs; Designing Training Programs; Role Specific and Competency Based Training; Evaluating Training Effectiveness; Performance appraisal: nature and objectives; Modern Techniques of performance appraisal;</td></tr>
<tr><td colspan="2">UNIT IV</td><td>7 hours</td></tr>
<tr><td colspan="3">Human Resource Development: Orientation Program; Requisite of an effective Program, Evaluation of Orientation Program. Strategic HRM: HRD audit, ethics and CSR</td></tr>
<tr><td colspan="3">Text Books</td></tr>
<tr><td>1</td><td colspan="2">G. Dessler. “A Framework for Human Resource Management”, Pearson Education, 2017, 15th Edition.</td></tr>
<tr><td>2</td><td colspan="2">D. A. Decenzo, S. P. Robbins, S. L. Verhulst, “Human Resource Management”, Wiley India Private Limited, 2015.</td></tr>
<tr><td colspan="3">Reference Books</td></tr>
<tr><td>1</td><td colspan="2">Bohlendar and Snell, “Principles of Human Resource Management”, Cengage Learning, 2013.</td></tr>
<tr><td>2</td><td colspan="2">B. Becker, M. Huselid, D. Ulrich, “The HR Scorecard”, 1st edition, Harvard Business Review Press, 2001.</td></tr>
</table>